

Decision Tree Classification.

Answer

The given dataset is

Machine	Temperature	Vibration	Failure
1	High	High	Yes
2	High	Low	Yes
3	Medium	High	Yes
4	Low	Low	No
5	Low	Medium	No
6	Medium	Low	No
7	High	Medium	Yes
8	Low	High	No

and it is given that, the new machine has

temperature = high
vibration = Medium.

From looking at the dataset, we can get that, for Failure.

Yes = 4

No = 4

So to calculate entropy,

$$\text{Entropy}(S) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$= - p_{\text{yes}} \log_2(p_{\text{yes}}) - p_{\text{no}} \log_2(p_{\text{no}})$$

$$= - \frac{4}{8} \log_2\left(\frac{4}{8}\right) - \frac{4}{8} \log_2\left(\frac{4}{8}\right)$$

$$= - 0.5 \log_2(0.5) - 0.5 \log_2(0.5)$$

$$= - 0.5 \log_2(0.5) - 0.5 \log_2(0.5) = 0.5 + 0.5 = 1$$

Weighted entropy for temperature is found as,

for high temperature, Yes = 3 No = 0.

$$\begin{aligned}\text{Entropy (s)} &= -P_{\text{yes}} \log_2(P_{\text{yes}}) - P_{\text{no}} \log_2(P_{\text{no}}) \\ &= -(1) \log_2(1) - 0 \log_2(0) \\ &= -1(0) \\ &= \underline{\underline{0}}\end{aligned}$$

for Medium temperature, Yes = 1 No = 1

$$\begin{aligned}\text{Entropy (s)} &= -P_{\text{yes}} \log_2(P_{\text{yes}}) - P_{\text{no}} \log_2(P_{\text{no}}) \\ &= -0.5 \log_2(0.5) - 0.5 \log_2(0.5) \\ &= -0.5(-1) - 0.5(-1) \\ &= \underline{\underline{1}}\end{aligned}$$

for low temperature, Yes = 0 No = 3

$$\begin{aligned}\text{Entropy (s)} &= -P_{\text{yes}} \log_2(P_{\text{yes}}) - P_{\text{no}} \log_2(P_{\text{no}}) \\ &= -(0) \log_2(0) - 1 \log_2(1) \\ &= -0 - 1(0) \\ &= \underline{\underline{0}}.\end{aligned}$$

$$\begin{aligned}\text{Weighted entropy (temp)} &= \frac{3}{8}(0) + \frac{2}{8}(1) + \frac{3}{8}(0) \\ &= \sum_{\text{VE values}} \frac{|S_v|}{|S|} \text{Entropy}(S_v) = \frac{2}{8} = \underline{\underline{0.25}}\end{aligned}$$

$$\begin{aligned}\text{Now Information gain for temperature} &= \text{Entropy} - \text{Weighted entropy} \\ &= 1 - 0.25 \\ &= \underline{\underline{0.75}}\end{aligned}$$

Weighted entropy for vibration is found as
for high vibration Yes = 2 No = 1

$$\begin{aligned}\text{Entropy} &= -P_{\text{yes}} \log_2(P_{\text{yes}}) - P_{\text{no}} \log_2(P_{\text{no}}) \\&= -\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right) \\&= -\frac{2}{3} (-0.585) - \frac{1}{3} (-1.585) \\&= \underline{\underline{0.918}}\end{aligned}$$

for medium vibration Yes = 1 No = 1

$$\begin{aligned}\text{Entropy} &= -0.5 \log_2(0.5) - 0.5 \log_2(0.5) \\&= \underline{\underline{1}}\end{aligned}$$

for low vibration Yes = 1 No = 2

$$\begin{aligned}\text{Entropy} &= -\frac{2}{3} \log_2\left(\frac{2}{3}\right) - \frac{1}{3} \log_2\left(\frac{1}{3}\right) - \frac{2}{3} \log_2\left(\frac{2}{3}\right) \\&= \underline{\underline{0.918}}\end{aligned}$$

$$\begin{aligned}\text{Weighted entropy (vibration)} &= \frac{3}{8} (0.918) + \frac{2}{8} (1) + \frac{3}{8} (0.918) \\&= 0.938\end{aligned}$$

Information Gain for Vibration

$$= \text{Entropy} - \text{Weighted entropy}$$

$$= 1 - 0.938$$

$$= 0.062$$

Now to find root node

Information Gain of temperature = 0.75

Information Gain of vibration = 0.062

Since information gain of temperature is higher we choose it as the root node

Now for the tree construction,

temperature = ~~Medium~~ High

All are yes

$\therefore$ leaf node = yes

High
|
Yes

temperature = Medium

Medium

Vibration
= high

|
Yes

Vibration
= Medium

|
Yes/No

Vibration = low

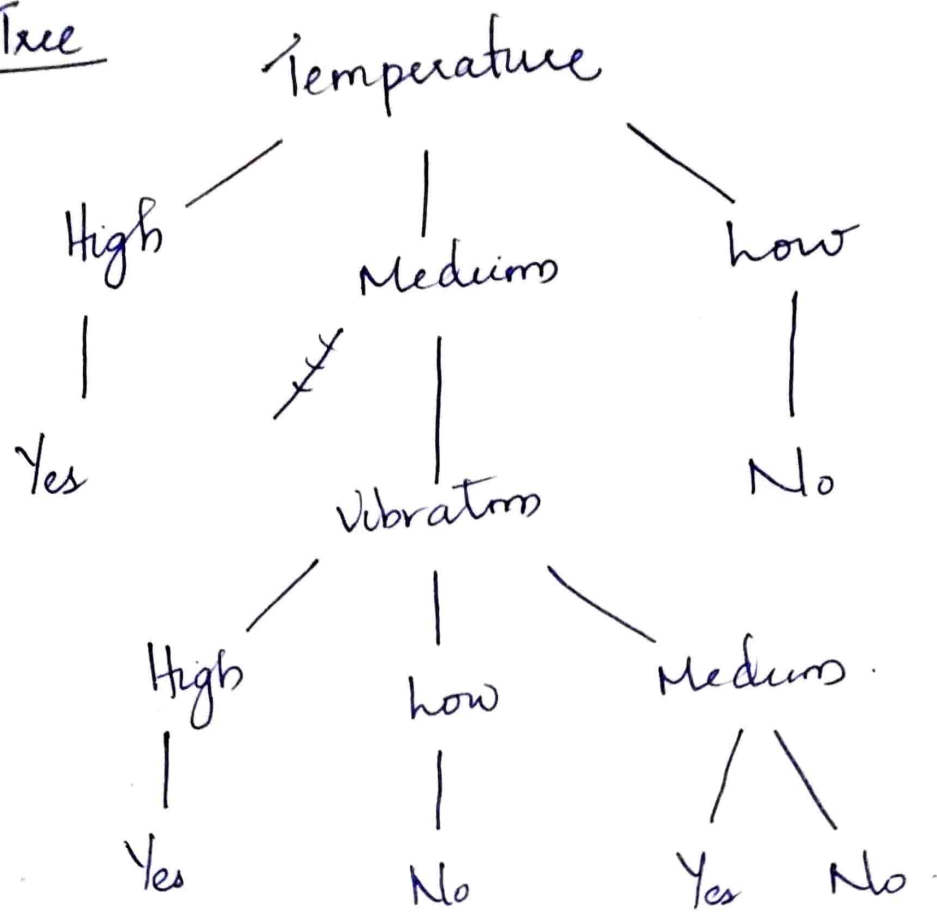
|
No

temperature = low

All are No

$\therefore$ leaf node = No

Tree



Now for the prediction,
the new machine has high temperature
and medium vibrations

Since the temperature is high the
changes of failure is Yes.